CAPSTONE PROJECT REPORT, TEAM ROOMORA

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Below is a report of the **Roomora** project as done by the **Roomora Team** listed above.

Project Definition

Roomora is simple a room booking website, where users are able to login, select the room(s) they wish to reserve including the duration with which they intend to stay and booking it for that duration. It was built using PHP, MySQL, HTML/CSS, and Bootstrap as instructed in the project brief.

Project Design Justification

We made use of property cards with clear images, prices and location to offer direct information to the user.

We made use of bootstrap for styling and making it as responsive as possible.

A blue get-started button which makes the call to action more visible as it contrasts with the white background.

Simple forms were use in the registration and login pages to make the website user friendly

The database schema we chose allows us to store the most relevant data about each property inserted into it. It is also easy to work with for the purpose of our website.

We have three separate tables in our schema to organize data collected; these are users, properties and bookings table.

Storage, Retrieval and Protection of Data

Data is **stored** using MySQL database named roomora, which contains three tables namely users, bookings and properties. All the details about the user collected via forms during register are stored in the users table. All the data about properties created when inserting them directly to the database are stored in the properties table and all the data about bookings collected when users book rooms are stored in the bookings table. For passwords, they are first hashed so that the hashed result can be stored, this is converting the password in plain text using a predefined algorithm to a unique sequence which can not be reversed to increase security. Additionally, each user record is assigned a unique ID number that automatically increases, making it easy to reference specific users in other parts of the system

Data **retrieval** is done in the following way:

When a user attempts to login, the email provided is matched to those stored in the users table during registration, if it exists and the password entered to the corresponding email after being hashed once more by the same hashing function matches the hashed password stored at the table then user is granted access to the account created for that email. Email matching is done by the query: SELECT * FROM users WHERE email = '$email.' A session is then created per each successfully authenticated user.

Data **protection** is done as follows:

To begin, storing the password in its hashed form and not converting it back to plaintext whenever user logs in enhances data security. Secondly, created sessions are stored on the server side to prevent manipulation by user. User data is also protected by keeping it specific to them – no other user can view or modify another user's data.

Basic Privacy Principles

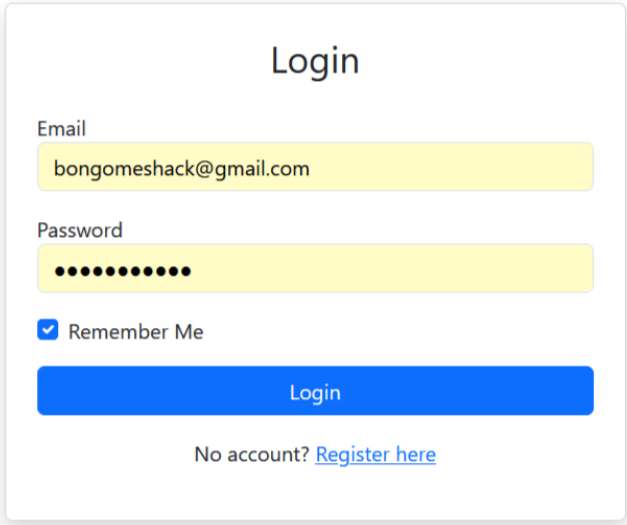
Roomora has adhered to the basic privacy principles in the following ways:

- Remember me feature is a check box which requires the user to opt in by clicking it. This ensures the system has the user's **consent** before creating a cookie for their data.
- User password is hashed enhancing **security**
- The system is transparent as user can clearly see what the "Remember Me" feature does; also, they can see if they are logged in.

Roles of Cookies in personalization and privacy risks

The Roomora system implements a "Remember Me" feature using cookies to enhance user experience.

When a user logs in and checks the "Remember Me" box, the system creates a persistent login session beyond the normal browser session. This will ensure user remains logged in even when they end the session. The cookie is deleted upon logging out.



In the above case cookies will serve to ensure the user remains logged in if they end session either accidentally or on purpose. This is beneficial for user experience and also convenient since they do not need to undergo the log in process every time they visit the website.

One major privacy risk is if the cookie is stolen, an attacker may also gain access to the user's account. The cookie is also susceptible to interception since we are not using HTTPS. In a public space like a cyber, users on public computers may remain logged in leading to potential data leaks.

Project Development Process

Planning and Design

By analyzing the capstone project brief, we concluded that our application must have the following core features:

- User Authentication.
- Responsive Design: Mobile-first layout using CSS Grid/Flexbox or Bootstrap.
- Database Operations: CRUD functionality for users, products/events/courses.
- Form Handling: Input validation (server-side and client-side).
- Cookies & Personalization: Store user preferences (e.g., theme, language, cart items).

We then decided to come up with a room booking application as it would allow us implement all the required core features and would later come to be known as **roomora** as suggested by Allan Njogu.

Following the data that would be collected and/or stored for the system, we came up with a simple database design framework that included 4 tables. The tables were:

- users – to store user data such as login info
- properties – to store details about the properties that would be in the application
- bookings – to store bookings made by users
- remember_tokens – for the "Remember Me" feature

We chose this schema because it's normalized, avoids data redundancy, and establishes clear relationships between tables.

Database Creation

Using a query in MySQL WorkBench, Allan Njogu created a database called roomora and moved on to creating the pre-mentioned tables above in the same query. Sample data was added to the properties table using the INSERT key word in the query.

Front End Design

Ian Mugi came up with the idea to have the assets folder which would hold a template for the header and footer of each page of the web application. These were made for the purpose of consistency across all pages. Bootstrap was used for its already predefined styles.

The login and registration forms were also styled with bootstrap.

Back End Design

We created config.php file for the database connection using mysqli() pre-defined function with the valid arguments.

For the user registration system, we implemented it with input validation and password hashing.

Login system was implemented with both email and password checking functionality; these would match the email and password from the form $POST method against that stored in the users data table.

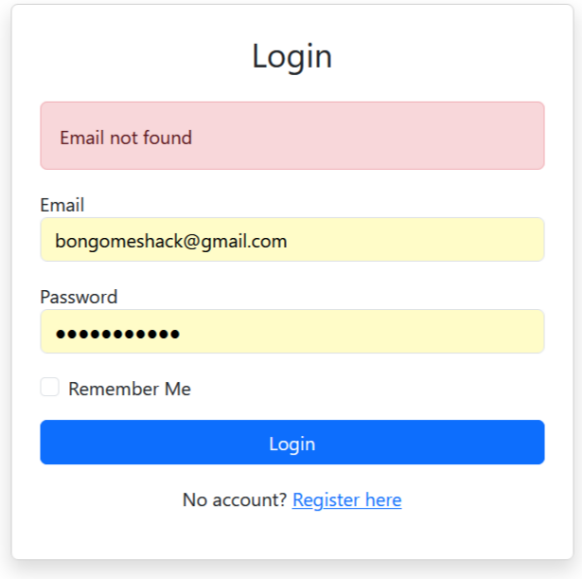
For displaying the properties, we referenced the database and looped through the properties table displaying dynamically.

For the booking logic; property-details.php retrieves the property ID from URL, checks date availability with an SQL query, calculates total price then adds it to booking table when confirmed.

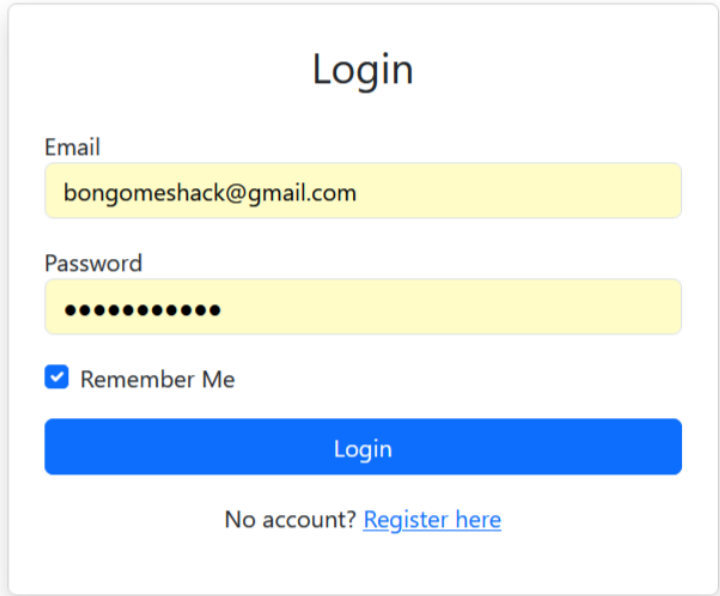
Deployment and Testing

We used the XAMPP control panel to host the site locally for the purpose of testing.

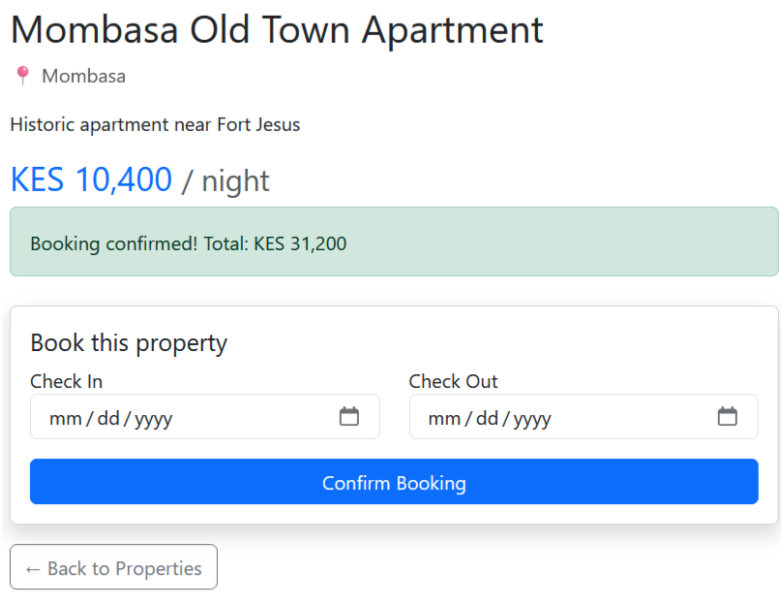
Once the web application was running, Bongo Sydney registered using the required details. Afterwards the wrong email was used to log in to check if the system knew how to handle wrong credentials.



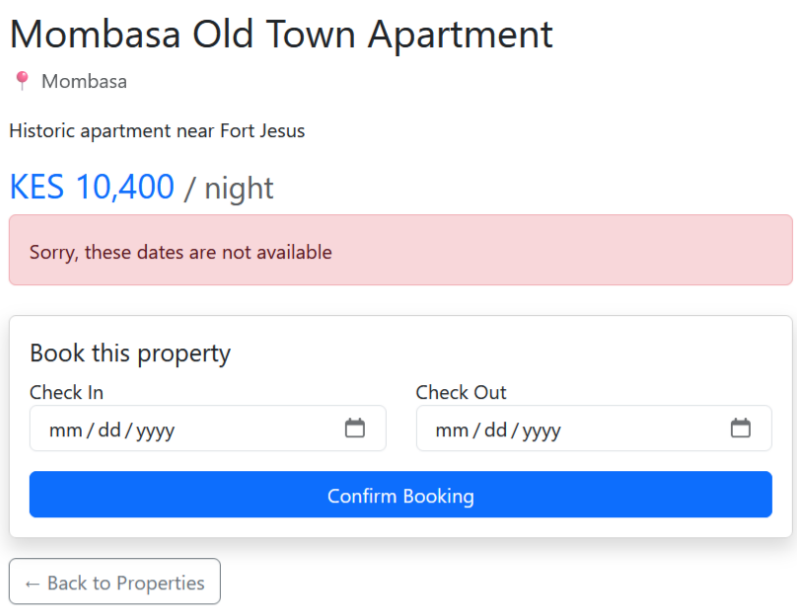
Afterwards the “Remember Me” feature was tested and we concluded that it worked after leaving the session and returning to a logged in account once we started a new ssession:



We moved on to testing the booking logic so as to check once booked there is an appropriate message displayed:



We tried checking if booking the same house on the same dates would display the appropriate message as well:



After doing all that we logged out to check whether the “Remember Me” feature would remain in effect. We were able to log out showing when you log out the cookie is destroyed as should be the case.

After extensive testing we concluded our project to meet the requirement specifications.

Team Roles

User authentication (Login, Register): Ian Mugi

Database creation: Allan Njogu

Booking logic: Bongo Sydney Meshack & Allan Njogu

Front End Design: Ian Mugi

Project Report: Bongo Sydney Meshack

Challenges and Solutions

Below comprises of the challenges we faced during the project development process as well as the respective solutions devised for each.

The first challenge of course was coming up with a lay out to the system. It comes to reason that in a team there will always be differing opinions, the solution tethered to this problem was that the final design should be user-friendly so as to raise our ranking and also interactive enough. We decided on having 4 optimal pages which meant the workload won't be as much for the simple website.

The second issue, was synchronization of our work. We decided it would best serve the cause if we divided the work and use a central point as a GitHub repository to bring it all together.

Evidence of Work

index.php:

```php
<?php include("includes/header.php");
require_once 'includes/config.php';
$properties = $db->query("SELECT * FROM properties LIMIT 3");
?>

<div class="container text-center mt-5">
    <h1 class="display-4">Welcome to Roomora</h1>
    <p class="lead">Book rooms easily and efficiently.</p>

    <div class="mt-4">
        <a href="register.php" class="btn btn-primary btn-lg me-3">Get Started</a>
        <a href="login.php" class="btn btn-outline-dark btn-lg">Login</a>
    </div>
</div>

<div class="container mt-5">
    <h3 class="text-center mb-4">Featured Properties</h3>
    <div class="row">
        <?php while ($prop = $properties->fetch_assoc()): ?>
            <div class="col-md-4 mb-3">
                <div class="card shadow">
                    <img src="<?php echo $prop['image_url']; ?>" class="card-img-top" style="height: 200px; object-fit: cover;">
                    <div class="card-body">
                        <h5><?php echo $prop['name']; ?></h5>
                        <p><?php echo $prop['location']; ?></p>
                        <p class="text-primary">KES <?php echo number_format($prop['price_per_night'] * 130); ?>/night</p>
                        <a href="property-details.php?id=<?php echo $prop['id']; ?>" class="btn btn-primary">View Details</a>
                    </div>
                </div>
            </div>
        <?php endwhile; ?>
    </div>
</div>
```

```php

<div class="container mt-5">
    <div class="row text-center">
        <div class="col-md-4">
            <div class="card shadow">
                <div class="card-body">
                    <h5>Easy Booking</h5>
                    <p>Book your desired room in seconds.</p>
                </div>
            </div>
        </div>

        <div class="col-md-4">
            <div class="card shadow">
                <div class="card-body">
                    <h5>Secure System</h5>
                    <p>Your data is safe and protected.</p>
                </div>
            </div>
        </div>

        <div class="col-md-4">
            <div class="card shadow">
                <div class="card-body">
                    <h5>Responsive Design</h5>
                    <p>Works perfectly on mobile & desktop.</p>
                </div>
            </div>
        </div>
    </div>
</div>

<?php include("includes/footer.php"); ?>
```

login.php:

```php
<?php
include("includes/header.php");

if (isset($_SESSION['user_id'])) {
    header("Location: dashboard.php");
    exit();
}

if (!isset($_SESSION['user_id']) && isset($_COOKIE['remember_token'])) {
    require_once 'includes/config.php';

    $token = $_COOKIE['remember_token'];
    $result = $db->query("SELECT u.* FROM users u JOIN remember_tokens rt ON u.id = rt.user_id WHERE rt.token = '$token' AND rt.expires_at > NOW()");

    if ($result->num_rows == 1) {
        $user = $result->fetch_assoc();
        $_SESSION['user_id'] = $user['id'];
        $_SESSION['user_name'] = $user['full_name'];
        header("Location: dashboard.php");
        exit();
    }
}

$error = '';
```

```php
if ($_SERVER['REQUEST_METHOD'] == 'POST') {
    $email = $_POST['email'];
    $password = $_POST['password'];
    $remember = isset($_POST['remember']) ? true : false;

    require_once 'includes/config.php';

    $result = $db->query("SELECT * FROM users WHERE email = '$email'");
    if ($result->num_rows == 1) {
        $user = $result->fetch_assoc();
        if (password_verify($password, $user['password'])) {
            $_SESSION['user_id'] = $user['id'];
            $_SESSION['user_name'] = $user['full_name'];

            if ($remember) {
                $token = bin2hex(random_bytes(32));
                $expires = date('Y-m-d H:i:s', strtotime('+30 days'));

                $db->query("DELETE FROM remember_tokens WHERE user_id = " . $user['id']);

                $db->query("INSERT INTO remember_tokens (user_id, token, expires_at) VALUES (" . $user['id'] . ", '$token', '$expires')");

                setcookie('remember_token', $token, [
                    'expires' => time() + (86400 * 30),
                    'path' => '/',
                    'domain' => '',
                    'secure' => false,
                    'httponly' => true,
                    'samesite' => 'Lax'
                ]);
            }

            header("Location: dashboard.php");
            exit();
        } else {
            $error = "Invalid password";
        }
    } else {
        $error = "Email not found";
    }
}
?>
```

```php
<div class="container mt-5" style="max-width: 500px;">
    <div class="card shadow p-4">
        <h3 class="text-center mb-4">Login</h3>

        <?php if ($error): ?>
            <div class="alert alert-danger"><?php echo $error; ?></div>
        <?php endif; ?>

        <form method="POST">
            <div class="mb-3">
                <label>Email</label>
                <input type="email" name="email" class="form-control" required>
            </div>

            <div class="mb-3">
                <label>Password</label>
                <input type="password" name="password" class="form-control" required>
            </div>

            <div class="mb-3 form-check">
                <input type="checkbox" name="remember" class="form-check-input" id="rememberCheck">
                <label class="form-check-label" for="rememberCheck">Remember Me</label>
            </div>

            <button type="submit" class="btn btn-primary w-100">Login</button>
        </form>

        <p class="text-center mt-3">
            No account? <a href="register.php">Register here</a>
        </p>
    </div>
</div>

<?php include("includes/footer.php"); ?>
```

logout.php:

```php
<?php
session_start();

require_once 'includes/config.php';

if (isset($_COOKIE['remember_token'])) {
    $token = mysqli_real_escape_string($db, $_COOKIE['remember_token']);

    $db->query("DELETE FROM remember_tokens WHERE token = '$token'");

    // deleting the cookie
    setcookie('remember_token', '', ['expires' => time() - 3600, 'path' => '/', 'domain' => '', 'secure' => false, 'httponly' => true,'samesite' => 'Lax'
    ]);
}

if (isset($_SESSION['user_id'])) {
    $user_id = $_SESSION['user_id'];
    $db->query("DELETE FROM remember_tokens WHERE user_id = $user_id");
}


$_SESSION = array();

if (ini_get("session.use_cookies")) {
    $params = session_get_cookie_params();
    setcookie(session_name(), '', time() - 42000, $params["path"], $params["domain"], $params["secure"], $params["httponly"]);
}

session_destroy();

header("Location: login.php");
exit();
```

Database schema:

```table_schema
CREATE TABLE properties (
    id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(200) NOT NULL,
    description TEXT,
    price_per_night DECIMAL(10,2) NOT NULL,
    location VARCHAR(255) NOT NULL,
    image_url VARCHAR(500),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

```sql
CREATE DATABASE IF NOT EXISTS roomora;
USE roomora;

CREATE TABLE users (
    id INT PRIMARY KEY AUTO_INCREMENT,
    full_name VARCHAR(100) NOT NULL,
    email VARCHAR(100) UNIQUE NOT NULL,
    password VARCHAR(255) NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE properties (
    id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(200) NOT NULL,
    description TEXT,
    price_per_night DECIMAL(10,2) NOT NULL,
    location VARCHAR(255) NOT NULL,
    image_url VARCHAR(500),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
CREATE TABLE bookings (
    id INT PRIMARY KEY AUTO_INCREMENT,
    user_id INT NOT NULL,
    property_id INT NOT NULL,
    check_in DATE NOT NULL,
    check_out DATE NOT NULL,
    total_price DECIMAL(10,2) NOT NULL,
    status ENUM('confirmed', 'cancelled') DEFAULT 'confirmed',
    booking_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE,
    FOREIGN KEY (property_id) REFERENCES properties(id) ON DELETE CASCADE
);
```

```sql
CREATE TABLE IF NOT EXISTS remember_tokens (
    id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT NOT NULL,
    token VARCHAR(64) NOT NULL,
    expires_at DATETIME NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE,
    UNIQUE KEY unique_token (token)
);

INSERT INTO properties (name, description, price_per_night, location, image_url) VALUES
('Diani Beach Villa', 'Beautiful beachfront villa with white sandy beaches', 150.00, 'D
('Mombasa Old Town Apartment', 'Historic apartment near Fort Jesus', 80.00, 'Mombasa',
('Malindi Seaview House', 'Relaxing house with ocean view near Marine Park', 120.00, 'M
```

register.php

```php
<?php
include("includes/header.php");

if (isset($_SESSION['user_id'])) {
    header("Location: dashboard.php");
    exit();
}

$error = '';
$success = '';

if ($_SERVER['REQUEST_METHOD'] == 'POST') {
    $name = $_POST['name'];
    $email = $_POST['email'];
    $password = $_POST['password'];
    $confirm = $_POST['confirm_password'];

    require_once 'includes/config.php';

    if ($password != $confirm) {
        $error = "Passwords do not match";
    } elseif (strlen($password) < 4) {
        $error = "Password too short";
    } else {
        $check = $db->query("SELECT id FROM users WHERE email = '$email'");
        if ($check->num_rows > 0) {
            $error = "Email already registered";
        } else {
            $hash = password_hash($password, PASSWORD_DEFAULT);
            $sql = "INSERT INTO users (full_name, email, password) VALUES ('$name', '$email', '$hash')";
            if ($db->query($sql)) {
                $success = "Registration successful! <a href='login.php'>Login here</a>";
            } else {
                $error = "Registration failed";
            }
        }
    }
}
?>
```

```php
<div class="container mt-5" style="max-width: 500px;">
    <div class="card shadow p-4">
        <h3 class="text-center mb-4">Sign Up</h3>

        <?php if ($error): ?>
            <div class="alert alert-danger"><?php echo $error; ?></div>
        <?php endif; ?>

        <?php if ($success): ?>
            <div class="alert alert-success"><?php echo $success; ?></div>
        <?php endif; ?>

        <form method="POST">
            <div class="mb-3">
                <label>Full Name</label>
                <input type="text" name="name" class="form-control" required>
            </div>

            <div class="mb-3">
                <label>Email</label>
                <input type="email" name="email" class="form-control" required>
            </div>

            <div class="mb-3">
                <label>Password</label>
                <input type="password" name="password" class="form-control" required>
            </div>

            <div class="mb-3">
                <label>Confirm Password</label>
                <input type="password" name="confirm_password" class="form-control" required>
            </div>

            <button type="submit" class="btn btn-primary w-100">Register</button>
        </form>

        <p class="text-center mt-3">
            Already have an account? <a href="login.php">Login here</a>
        </p>
    </div>
</div>

<?php include("includes/footer.php"); ?>
```

assets:

style.css:

```css
html,
body {
    height: 100%;
    margin: 0;
    padding: 0;
}

body {
    display: flex;
    flex-direction: column;
    min-height: 100vh;
}

.container {
    flex: 1 0 auto;
}

footer {
    flex-shrink: 0;
    width: 100%;
}
```

includes(files)

config.php:

```php
<?php

$host = 'localhost';
$dbname = 'roomora';
$username = 'root';
$password = '@bongo4life';

$db = new mysqli($host, $username, $password, $dbname);


if ($db->connect_error) {
    die("Connection failed: " . $db->connect_error);
}
```

header.php:

```php
<?php
if (session_status() === PHP_SESSION_NONE) {
    session_start();
}
?>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Roomora</title>
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
    <link rel="stylesheet" href="assets/css/style.css">
</head>
<body>
    <nav class="navbar navbar-expand-lg navbar-dark bg-dark">
        <div class="container">
            <a class="navbar-brand" href="index.php">Roomora</a>

            <div class="ms-auto">
                <?php if (isset($_SESSION['user_name'])): ?>

                    <span class="text-white me-3">
                        Hi, <?php echo $_SESSION['user_name']; ?>
                    </span>

                    <a href="logout.php" class="btn btn-danger btn-sm">
                        Logout
                    </a>

                <?php else: ?>

                    <a href="login.php" class="btn btn-outline-light btn-sm me-2">
                        Login
                    </a>
                    <a href="register.php" class="btn btn-primary btn-sm">
                        Sign Up
                    </a>

                <?php endif; ?>
            </div>
        </div>
    </nav>
```

footer.php:

```
<footer class="bg-dark text-white text-center p-3 mt-5">
    <p class="mb-0">© <?php echo date("Y"); ?> Roomora. All rights reserved.</p>
</footer>

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
<script src="assets/js/script.js"></script>

</body>

</html>
```

dashboard.php:

```
<?php
include("includes/header.php");
if (!isset($_SESSION['user_id'])) {
    header("Location: login.php");
    exit();
}
require_once 'includes/config.php';
$user_id = $_SESSION['user_id'];
$bookings_result = $db->query("SELECT COUNT(*) as total FROM bookings WHERE user_id = $user_id");
$bookings_count = $bookings_result->fetch_assoc()['total'];
$properties_result = $db->query("SELECT COUNT(*) as total FROM properties");
$properties_count = $properties_result->fetch_assoc()['total'];
?>
<div class="container mt-5">
    <h2>Welcome, <?php echo $_SESSION['user_name']; ?> 👋. Nice to have you back!</h2>
    <p id="clock" class="text-muted"></p>
    <div class="row mt-4">
        <div class="col-md-4 mb-3">
            <div class="card shadow text-center p-3">
                <h5>My Bookings</h5>
                <h3><?php echo $bookings_count; ?></h3>
                <a href="my-bookings.php" class="btn btn-sm btn-outline-primary mt-2">View</a>
            </div>
        </div>
        <div class="col-md-4 mb-3">
            <div class="card shadow text-center p-3">
                <h5>Available Rooms</h5>
                <h3><?php echo $properties_count; ?></h3>
                <a href="index.php" class="btn btn-sm btn-outline-primary mt-2">Browse</a>
            </div>
        </div>
        <div class="col-md-4 mb-3">
            <div class="card shadow text-center p-3">
                <h5>Upcoming Reservations</h5>
                <h3><?php echo $bookings_count; ?></h3>
            </div>
        </div>
    </div>
</div>

<?php include("includes/footer.php"); ?>
```

my-bookings.php:

```php
<?php
include("includes/header.php");

if (!isset($_SESSION['user_id'])) {
    header("Location: login.php");
    exit();
}

require_once 'includes/config.php';
$user_id = $_SESSION['user_id'];

$bookings = $db->query("SELECT b.*, p.name, p.location, p.image_url
                        FROM bookings b
                        JOIN properties p ON b.property_id = p.id
                        WHERE b.user_id = $user_id
                        ORDER BY b.booking_date DESC");
?>
```

```php
<div class="container mt-5">
    <h2>My Bookings</h2>

    <?php if ($bookings->num_rows == 0): ?>
        <div class="alert alert-info">You have no bookings yet.</div>
    <?php else: ?>
        <div class="row">
            <?php while ($booking = $bookings->fetch_assoc()): ?>
                <div class="col-md-6 mb-3">
                    <div class="card shadow">
                        <div class="row g-0">
                            <div class="col-md-4">
                                <img src="<?php echo $booking['image_url']; ?>" class="img-fluid rounded-start" style="height: 100%; object-fit: cover;">
                            </div>
                            <div class="col-md-8">
                                <div class="card-body">
                                    <h5><?php echo $booking['name']; ?></h5>
                                    <p><?php echo $booking['location']; ?></p>
                                    <p>Check in: <?php echo $booking['check_in']; ?></p>
                                    <p>Check out: <?php echo $booking['check_out']; ?></p>
                                    <p class="text-primary">Total: KES <?php echo number_format($booking['total_price'] * 130); ?></p>
                                    <span class="badge bg-success">&lt;?php echo $booking['status']; ?&gt;</span>
                                </div>
                            </div>
                        </div>
                    </div>
                </div>
            <?php endwhile; ?>
        </div>
    <?php endif; ?>

    <a href="dashboard.php" class="btn btn-primary mt-3">Back to Dashboard</a>
</div>

<?php include("includes/footer.php"); ?>
```

property-details.php:

```php
<?php
include("includes/header.php");
if (!isset($_GET['id'])) {
    header("Location: index.php");
    exit();
}
$id = $_GET['id'];
require_once 'includes/config.php';
$property = $db->query("SELECT * FROM properties WHERE id = $id")->fetch_assoc();

if (!$property) {
    header("Location: index.php");
    exit();
}
$error = '';
$success = '';
if ($_SERVER['REQUEST_METHOD'] == 'POST' && isset($_SESSION['user_id'])) {
    $check_in = $_POST['check_in'];
    $check_out = $_POST['check_out'];
    $user_id = $_SESSION['user_id'];

    $check = $db->query("SELECT * FROM bookings WHERE property_id = $id
                         AND status='confirmed'
                         AND ((check_in BETWEEN '$check_in' AND '$check_out')
                         OR (check_out BETWEEN '$check_in' AND '$check_out'))");

    if ($check->num_rows > 0) {
        $error = "Sorry, these dates are not available";
    } else {
        $days = (strtotime($check_out) - strtotime($check_in)) / (60 * 60 * 24);
        $total = $days * $property['price_per_night'];

        $sql = "INSERT INTO bookings (user_id, property_id, check_in, check_out, total_price)
                VALUES ($user_id, $id, '$check_in', '$check_out', $total)";

        if ($db->query($sql)) {
            $success = "Booking confirmed! Total: KES " . number_format($total * 130);
        } else {
            $error = "Booking failed. Please try again.";
        }
    }
}
?>
```

```php
<div class="container mt-5">
    <div class="row">
        <div class="col-md-6">
            <img src="<?php echo $property['image_url']; ?>" class="img-fluid rounded shadow" alt="<?php echo $property['name']; ?>">
        </div>
        <div class="col-md-6">
            <h2><?php echo $property['name']; ?></h2>
            <p class="text-muted">📍 <?php echo $property['location']; ?></p>
            <p><?php echo $property['description']; ?></p>
            <h3 class="text-primary">KES <?php echo number_format($property['price_per_night'] * 130); ?> <small class="text-muted">/ night</small></h3>
            <?php if ($error): ?>
                <div class="alert alert-danger"><?php echo $error; ?></div>
            <?php endif; ?>
            <?php if ($success): ?>
                <div class="alert alert-success"><?php echo $success; ?></div>
            <?php endif; ?>
            <?php if (isset($_SESSION['user_id])): ?>
                <div class="card shadow p-3 mt-4">
                    <h5>Book this property</h5>
                    <form method="POST">
                        <div class="row">
                            <div class="col-md-6 mb-3">
                                <label>Check In</label>
                                <input type="date" name="check_in" class="form-control" required>
                            </div>
                            <div class="col-md-6 mb-3">
                                <label>Check Out</label>
                                <input type="date" name="check_out" class="form-control" required>
                            </div>
                        </div>
                        <button type="submit" class="btn btn-primary w-100">Confirm Booking</button>
                    </form>
                </div>
            <?php else: ?>
                <div class="alert alert-info mt-3">
                    Please <a href="login.php">login</a> to book this property.
                </div>
            <?php endif; ?>
            <a href="index.php" class="btn btn-outline-secondary mt-3">← Back to Properties</a>
        </div>
    </div>
</div>
<?php include("includes/footer.php"); ?>
```

Hosting and SEO plan

Current plans of hosting include use of a local server, XAMPP Apache to be precise. This is both free, easily available and easy to test and debug.

Future plans of hosting include use of free web hosting services such as AwardSpace which is good for dynamic websites as these.

To deploy the website, I would need to:

- Start by uploading the code files on the created GitHub repository
- Link the GitHub associated with the repository to a AwardSpace account
- Export my database to an SQL file
- On AwardSpace, locate the deployment option and follow instructions given.

The following are SEO techniques applied to raise the websites ranking on searches.

- Use of metatags in header.php to give web page descriptions to the user.
- The website was optimized to also be responsive for mobile users thus raise ranking in mobile searches.
- We used minimal CSS and JS to ensure the page loads as fast as possible with 4G Wi-Fi

The following are suggested future Seo improvements.

- Adding metatags for each page so as to enable each user understand purpose of every page.
- Add proper billing system to users who want to pay
- Add a cancelation feature for reserved rooms 3 days before the reserved start duration
- Use Cleaner URLs